INCH-POUND

MIL-C-83526/4B
14 Oct 1993
SUPERSEDING
MIL-C-83526/4A
30 June 1989

MILITARY SPECIFICATION SHEET

# CONNECTORS, RECEPTACLE, FIBER OPTIC, CIRCULAR, HERMAPHRODITIC, JAM-NUT MOUNTING, 6 POSITIONS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83526.

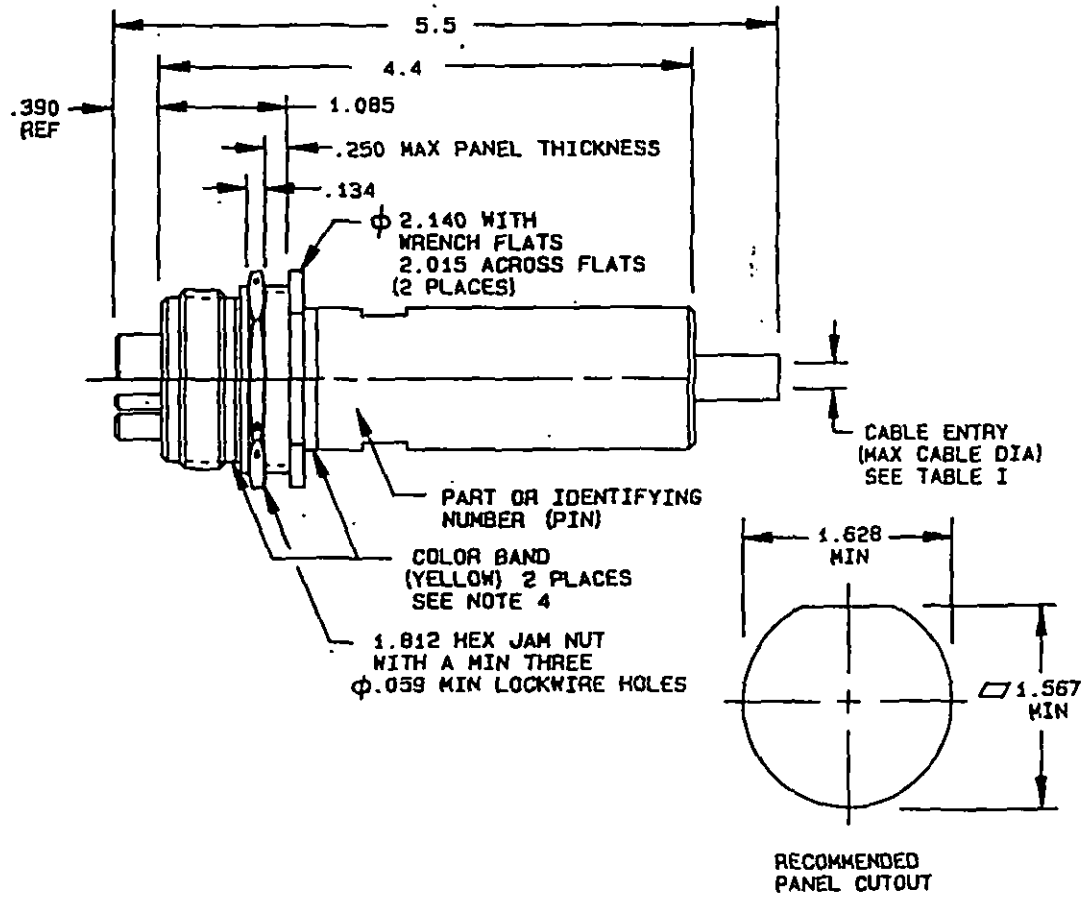
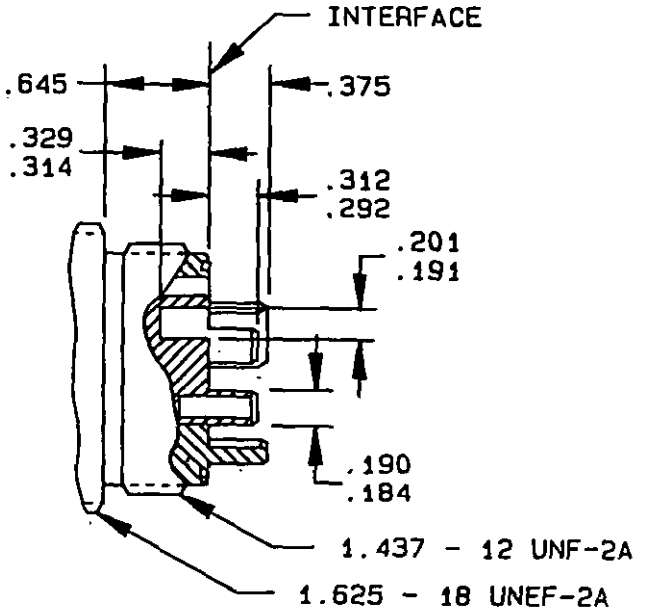



FIGURE 1. Jam-nut connector.


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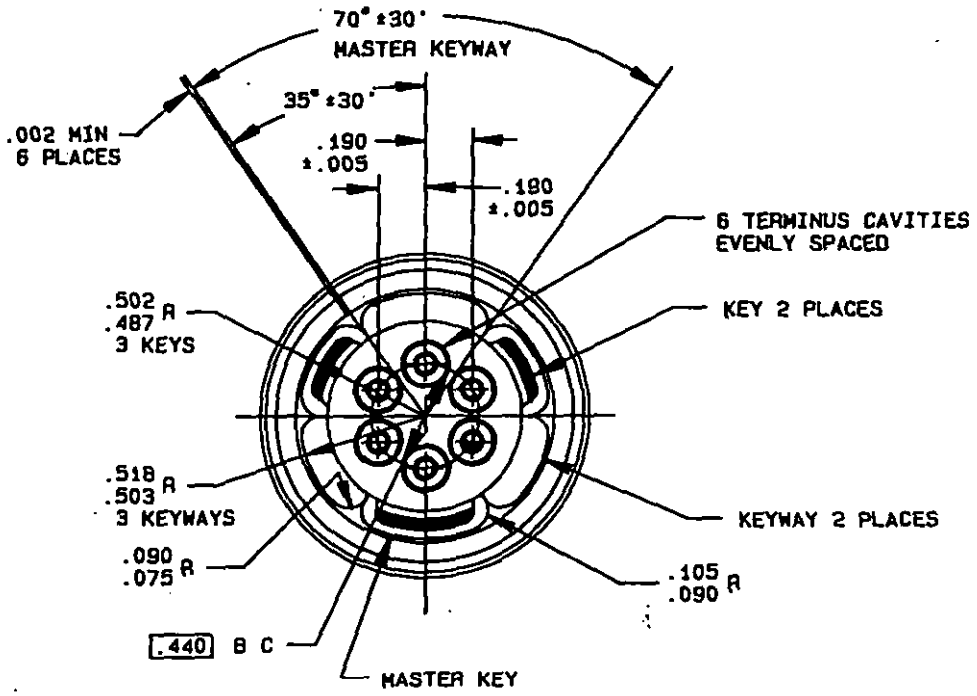

| Inches | mm | Inches | mm |
|---|---|---|---|
| .059 | 1.50 | .390 | 9.91 |
| .134 | 3.40 | .645 | 16.38 |
| .184 | 4.67 | 1.085 | 27.56 |
| .190 | 4.83 | 1.437 | 36.50 |
| .191 | 4.85 | 1.567 | 39.80 |
| .201 | 5.11 | 1.625 | 41.28 |
| .250 | 6.35 | 1.628 | 41.35 |
| .292 | 7.42 | 1.812 | 46.02 |
| .312 | 7.92 | 2.015 | 51.18 |
| .314 | 7.98 | 2.140 | 54.36 |
| .329 | 8.36 | 4.4 | 112 |
| .375 | 9.52 | 5.5 | 140 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± 0.010 (0.25 mm) for three place decimals and ± 0.03 (0.8 mm) for two place decimals, and ± 0.1 (3 mm) for one place decimals. Angular tolerances are ± 0° 30'.
4. The phrase "Fiber-Optic" must be included within the yellow band.
5. Unless otherwise specified, dimensions are symmetrical about centerlines.

FIGURE 1. Jam-nut connector, continued.



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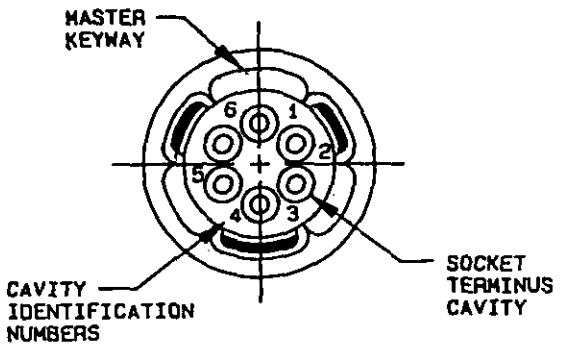

| Inches | mm |
|---|---|
| .002 | 0.05 |
| .005 | 0.13 |
| .075 | 1.90 |
| .090 | 2.29 |
| .105 | 2.67 |
| .140 | 3.56 |
| .440 | 11.18 |
| .487 | 12.37 |
| .502 | 12.75 |
| .503 | 12.78 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

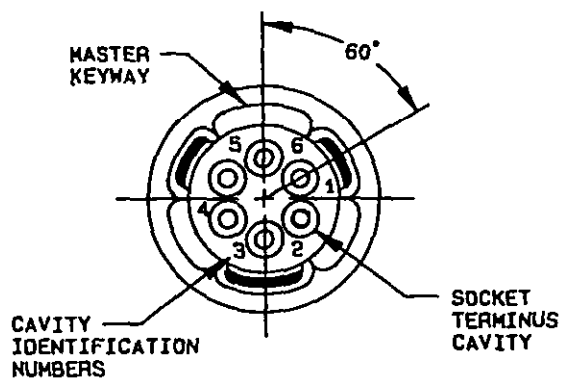
FIGURE 2. Terminal configuration face view.



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KEYING POSITION A




KEYING POSITION B

NOTE: Cavities 1, 3, and 5 are socket terminus cavities.

FIGURE 3. Keying positions.



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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figures 1, 2, and 3.

Fiber optic cable:

Cable diameter: In accordance with MIL-C-85045.

Cable configuration: In accordance with MIL-C-85045.

Fiber diameter: In accordance with MIL-F-49291.

Fiber numerical aperture: In accordance with MIL-F-49291.

Fiber type: Multimode.

Fiber class: Graded index.

Fiber composition: Glass/glass.

Materials: The material requirements are listed in table I.

Connector termination and assembly tools: As specified in MIL-K-83533.

Cleaning procedures: The cleaning procedures for the connector shall be included with the assembly instructions of the connector.

Termini: The required number of termini shall be supplied with each connector unless otherwise specified. Style P(pin) or S(socket), see MIL-T-29504/1 and MIL-T-29504/2. For dummy terminus, see MIL-T-29504/3.

Weight: 9.0 ounces maximum.

Coupling forces: 11.0 pounds.

Coupling torques: 5 inch-pounds minimum.



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TABLE I. Material description.

| Parts description | Material | Finish |
|---|---|---|
| Connector shell<br>Insert (6 termini) | Aluminum alloy<br>2011-T3<br>QQ-A-225/3 | Anodized<br>MIL-A-8625, type II<br>class 2, Color: gray |
| Insert retaining nut<br>Spacer<br>Load bracket<br>Protecting sleeve<br>Locking nut<br>Jam nut<br>Inner sleeve | Aluminum alloy<br>2011-T3<br>QQ-A-225/3 | Chemical film<br>MIL-C-5541,<br>class 1A<br>"<br>"<br>"<br>" |
| Shaft retaining screw<br>Dowel pin | CRES<br>" | Passivated<br>MIL-S-5002 |
| Cable retaining shaft | Beryllium Copper<br>QQ-C-530 | None |
| Shrink tubing | Radiation crosslinked<br>Polyolefin<br>MIL-I-23053/4, class 3 | None<br>None |
| O-ring | Fluorosilicone rubber<br>MIL-R-25988, class 1<br>grade 70 | None |

PIN:

M83526/04 X X X X

Basic specification sheet

Number of terminus positions required

| Quantity | Designator |
|---|---|
| 6 | C |

Cable size 1/

| Cable OD in inches | Designator |
|---|---|
| 0.217 - 0.256 | 2 |

Terminus supplied for actual fiber size 2/

| Fiber OD in microns | Designator |
|---|---|
| 123.0 - 124.4 | A |
| 124.5 - 127.0 | B |
| 138.0 - 139.6 | C |
| 139.7 - 142.0 | D |
| No termini supplied | E |

Keying position of terminus insert arrangement

| Position | Designator |
|---|---|
| A (STD) | 1 |
| (B) Rotated 60º | 2 |

1/ Select cable size and type in accordance with MIL-C-85045.
2/ Select fiber sizes and tolerances in accordance with MIL-F-49291.



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Quality conformance and periodic inspection: The following cables shall be used for quality conformance and periodic inspection.

| Number channels | Cables |
|---|---|
| 6 | PIN M85045/2C6A, 100 micron, 140 clad, or equal. |

For quality conformance and periodic inspection, dummy termini shall be used in all unused cavities.

Patent notice: The Government does not have a royalty-free license under the following listed patents for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

| Patent number | Patent expiration date |
|---|---|
| US 3,994,564 | 11/30/1993 |
| US 4,054,366 | 10/18/1994 |
| US 4,060,308 | 11/29/1994 |

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - MI
Navy - AS, SH
Air Force - 17, 80
DLA - ES

User activities:
Navy - MC, OS, YD
Air Force - 13

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6060-0100-04)



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